Recurrent Neural Network

Tadagbé Dhossou

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1 Introduction

Recurrent Neural Networks (RNNs) is a specific kind of neural networks that process sequential data by maintainng a hidden state that acts as memory. At each timestep, the RNN update its hidden state by considering both the current input and the past hidden state. This mechanisms help the model to capture temporal dependencies in sequential data.

2 Forward Pass

Consider an input sequence

$$x = (x_1, x_2, x_3, ..., x_T)$$

where T denotes the length of the input sequence and x_t denotes the input passed to the model at time step t. The RNN processes the sequence step by step while maintaining a hidden state h_t at each time step t.

The hidden state update is defined by

$$h_t = \tanh(W_{xh}x_t + W_{hh}h_{t-1} + b_h)$$

- W_{xh} is the weight matrix connecting the input to the hidden state
- W_{hh} is the hidden matrix connecting the current hidden state to the past hidden state
- b_h is the hidden state bias vector
- $\tanh(\cdot)$ is an activation function that is applied element-wise. It squashes the input value into the range $[-1, 1]$.

The hidden state h_t acts like a memory that maintain the information related to all the previous time step. The output of the neural network at time step t is computed by

$$y_t = W_{hy}h_t + b_y$$

where

- W_{hy} is the weight connecting the hidden state to the output
- b_y is the output bias vector

In our case, we will consider a particular case of applicaiton of our RNN where we want our model to predict the next character. This can be seen as a classification problem where at each time step we want the model to predict which character is more likely to appear next. It is a classification problem because to achieve this, we just need to predict a probabibility distribution over all the character of our alpahabet. The one that our model will consider to be the more likely to appear next is the one that it will attribute the highest probability. To accomplish this, we need to convert our y_t vector in probabilities. We can use the softmax function for this purpose.

$$p_t = \text{softmax}(y_t)$$

3 Loss Function

The loss at each time step t is defined by:

$$L_t = -\sum_{i=1}^{V} y_t^{true}[i] \log(p_t[i])$$

The loss for a sequence

$$x = (x_1, x_2, x_3, ..., x_T)$$

the total loss is then:

$$L = \sum_{t=1}^{T} L_t$$

4 Backpropagation

4.1 Backpropagation at each time step

The parameters of the network

$$\theta = \{W_{xh}, W_{hh}, W_{hy}, b_h, b_y\}$$

are learned during training using gradient-based optimization methods.

At each time step t, the RNN performs the following computation:

$$z_t = W_{xh}x_t + W_{hh}h_{t-1} + b_h \tag{1}$$
$$h_t = \tanh(z_t) \tag{2}$$
$$y_t = W_{hy}h_t + b_y \tag{3}$$
$$p_t = \text{softmax}(y_t) \tag{4}$$
$$L_t = -\sum_{i=1}^{V} y_t^{true}[i] \log(p_t[i]) \tag{5}$$
$$L = \sum_{t=1}^{T} L_t \tag{6}$$

4.1.1 Dimensions

Let:

$$V = \text{vocab_size}, \quad H = \text{hidden_size}$$

$$x_t \in R^{V \times 1}$$

$$z_t \in R^{H \times 1}$$

$$h_t \in R^{H \times 1}$$

$$y_t \in R^{V \times 1}$$

$$p_t \in R^{V \times 1}$$

$$W_{xh} \in R^{H \times V}$$

$$W_{hh} \in R^{H \times H}$$

$$W_{hy} \in R^{V \times H}$$

$$b_h \in R^{H \times 1}$$

$$b_y \in R^{V \times 1}$$

4.1.2 Visualization

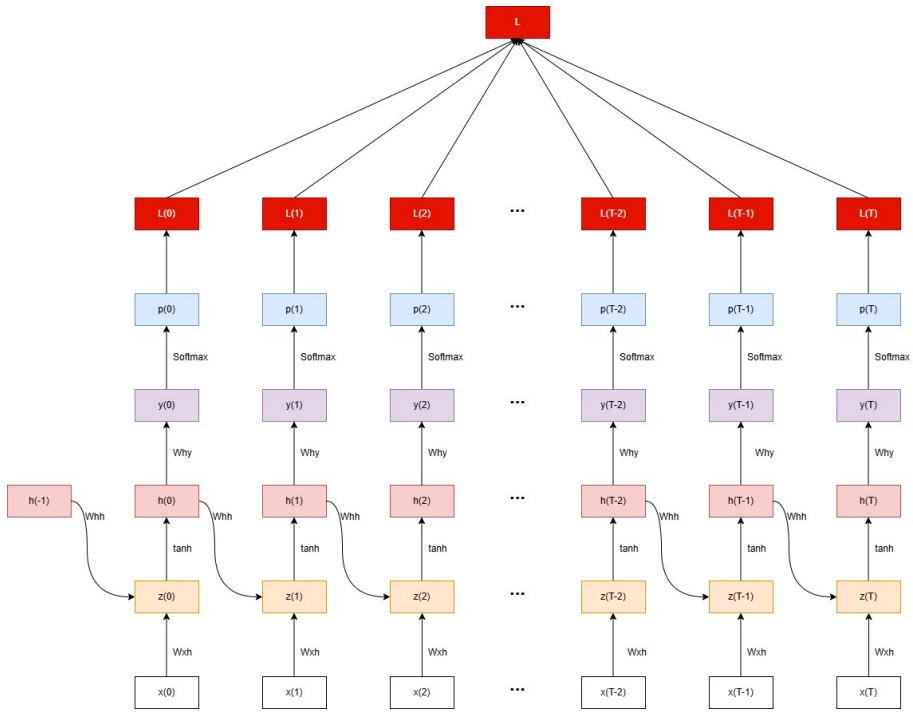

Figure 1: Unrolled Recurrent Neural Network from time step 1 to T

4.1.3 Intermediate steps

$$\frac{\partial L_t}{\partial p_t[j]} = -\frac{y_t^{\text{true}}[j]}{p_t[j]}$$

$$\frac{\partial L_t}{\partial y_t} = \frac{\partial L_t}{\partial p_t}\frac{\partial p_t}{\partial y_t}$$

For all j

$$\frac{\partial L_t}{\partial y_t[j]} = \sum_i \frac{\partial L_t}{\partial p_t[i]}\frac{\partial p_t[i]}{\partial y_t[j]}$$

In a precedent results, we got:

$$\frac{\partial L_t}{\partial p_t[i]} = -\frac{y_t^{\text{true}}[i]}{p_t[i]}$$

So for all j:

$$\frac{\partial L_t}{\partial y_t[j]} = \sum_i -\frac{y_t^{\text{true}}[i]}{p_t[i]}\frac{\partial p_t[i]}{\partial y_t[j]}$$

y_t^{true} is a one-hot encoding vector so contains a non-nul value a only one position, and this value is equal to one (1). Let's denote by k this position.

$$\frac{\partial L_t}{\partial y_t[j]} = -\frac{y_t^{\text{true}}[k]}{p_t[k]} \frac{\partial p_t[k]}{\partial y_t[j]}$$

$$p_t = \text{softmax}(y_t)$$

$$p_t[k] = \frac{e^{y_t[k]}}{\sum_l e^{y_t[l]}}$$

$$\frac{\partial p_t[k]}{\partial y_t[j]} = \frac{\frac{\partial e^{y_t[k]}}{\partial y_t[j]} \sum_l e^{y_t[l]} - \frac{\partial \sum_l e^{y_t[l]}}{\partial y_t[j]} e^{y_t[k]}}{(\sum_l e^{y_t[l]})^2}$$

$$\frac{\partial p_t[k]}{\partial y_t[j]} = \frac{\frac{\partial e^{y_t[k]}}{\partial y_t[j]} \sum_l e^{y_t[l]} - e^{y_t[j]} e^{y_t[k]}}{(\sum_l e^{y_t[l]})^2}$$

$$\frac{\partial p_t[k]}{\partial y_t[j]} = \frac{\delta_{kj} e^{y_t[k]} \sum_l e^{y_t[l]} - e^{y_t[j]} e^{y_t[k]}}{(\sum_l e^{y_t[l]})^2}$$

$$\frac{\partial p_t[k]}{\partial y_t[j]} = \delta_{kj} p_t[k] - p_t[j] p_t[k]$$

$$\frac{\partial L_t}{\partial y_t[j]} = -\frac{1}{p_t[k]} (\delta_{kj} p_t[k] - p_t[j] p_t[k])$$

$$\frac{\partial L_t}{\partial y_t[j]} = p_t[j] - \delta_{kj}$$

We can valablely replace δ_{kj} by $y_t^{\text{true}}[j]$. Why ??
Hint: Think about the one hot encoding (only 0 and 1; and the value of y_t^{true} at position k and j when k and j are equal or not.

Then we obtain:

$$\frac{\partial L_t}{\partial y_t[j]} = p_t[j] - y_t^{\text{true}}[j]$$

We can then conclude that

$$\frac{\partial L_t}{\partial y_t} = p_t - y_t^{\text{true}}$$

4.1.4 Calculation of $\frac{\partial L}{\partial L_t}$

L_t is going to influence L only through L_t. So

$$\frac{\partial L}{\partial L_t} = 1$$

4.1.5 Calculation of $\frac{\partial L}{\partial p_t}$

p_t is going to influence L only through L_t. So

$$\frac{\partial L}{\partial p_t[i]} = \frac{\partial L_t}{\partial p_t[i]}$$

$$\begin{aligned}\frac{\partial L}{\partial p_t[i]} &= \frac{\partial - \sum_{i=1}^{V} y_t^{true}[i] \log(p_t[i])}{\partial p_t[i]} \\ &= \frac{-y_t^{true}[i]}{p_t[i]}\end{aligned}$$

$$\boxed{\frac{\partial L}{\partial p_t} = \frac{-y_t^{true}}{p_t}}$$

4.1.6 Calculation of $\frac{\partial L}{\partial y_t}$

$$\frac{\partial L}{\partial y_t[i]} = \frac{\partial L_t}{\partial y_t[i]}$$

$$\begin{aligned}\frac{\partial L}{\partial y_t[i]} &= \frac{\partial L_t}{\partial y_t[i]} \\ &= \sum_{j=1}^{V} \frac{\partial L_t}{\partial p_t[j]} \frac{\partial p_t[j]}{\partial y_t[i]} \quad \text{because a small change in } y_t[i] \text{ is going to affect all the elements in } p_t \\ &= p_t[i] - y_t^{true}[i]\end{aligned}$$

$$\boxed{\frac{\partial L}{\partial y_t} = p_t - y_t^{true}}$$

Let's pose

$$\delta_t^y = \frac{\partial L}{\partial y_t} = p_t - y_t^{true}$$

4.1.7 Calculation of $\frac{\partial L}{\partial W_{hy}}$

$$\begin{aligned}\frac{\partial L}{\partial W_{hy}[i,j]} &= \sum_{t=1}^{T} \frac{\partial L_t}{\partial W_{hy}[i,j]} \quad \text{because } W_{hy} \text{ is going to influence } L \text{ through each } L_t \\ &= \sum_{t=1}^{T} \frac{\partial L_t}{\partial y_t[i]} \frac{\partial y_t[i]}{\partial W_{hy}[i,j]} \\ &= \sum_{t=1}^{T} \left(p_t[i] - y_t^{true}[i]\right) h_t[j]\end{aligned}$$

$$\boxed{\frac{\partial L}{\partial W_{hy}} = \sum_{t=1}^{T} \delta_t^y h_t^T}$$

4.1.8 Calculation of $\frac{\partial L}{\partial h_t}$

We distinguish two cases.

Case 1: $t = T$ At the final time step, $h_T[i]$ influences the total loss only through L_T. Therefore,

$$\begin{aligned}
\frac{\partial L}{\partial h_T[i]} &= \frac{\partial L_T}{\partial h_T[i]} \\
&= \sum_{j=1}^{V} \frac{\partial L_T}{\partial y_T[j]} \frac{\partial y_T[j]}{\partial h_T[i]} \\
&= \sum_{j=1}^{V} \left(p_T[j] - y_T^{\text{true}}[j]\right) W_{hy}[j,i].
\end{aligned}$$

Case 2: $t < T$ For $t < T$, $h_t[i]$ influences the total loss in two ways:

- directly through L_t via y_t,
- indirectly through future time steps via z_{t+1}.

Thus,

$$\begin{aligned}
\frac{\partial L}{\partial h_t[i]} &= \sum_{j=1}^{V} \frac{\partial L_t}{\partial y_t[j]} \frac{\partial y_t[j]}{\partial h_t[i]} + \sum_{k=1}^{H} \frac{\partial L}{\partial z_{t+1}[k]} \frac{\partial z_{t+1}[k]}{\partial h_t[i]} \\
&= \sum_{j=1}^{V} \left(p_t[j] - y_t^{\text{true}}[j]\right) W_{hy}[j,i] + \sum_{k=1}^{H} \frac{\partial L}{\partial z_{t+1}[k]} W_{hh}[k,i].
\end{aligned}$$

Let's pose $\delta_t = \frac{\partial L}{\partial z_t}$

Case 1: $t < T$

$$\boxed{\frac{\partial L}{\partial h_t} = W_{hy}^T \delta_t^y + W_{hh}^T \delta_{t+1}}$$

Case 2: $t = T$

$$\boxed{\frac{\partial L}{\partial h_T} = W_{hy}^T \delta_T^y}$$

4.1.9 Calculation of $\frac{\partial L}{\partial z_t}$

$$\begin{aligned}
\frac{\partial L}{\partial z_t[i]} &= \frac{\partial L}{\partial h_t[i]} \frac{\partial h_t[i]}{\partial z_t[i]} \\
&= \frac{\partial L}{\partial h_t[i]} \left(1 - \tanh^2(z_t[i])\right) \\
&= \frac{\partial L}{\partial h_t[i]} \left(1 - h_t[i]^2)\right)
\end{aligned}$$

$$\boxed{\frac{\partial L}{\partial z_t} = \frac{\partial L}{\partial h_t} \odot (1 - h_t \odot h_t)}$$

4.1.10 Calculation of $\frac{\partial L}{\partial W_{xh}}$

$$\begin{aligned}
\frac{\partial L}{\partial W_{xh}[i,j]} &= \sum_{t=1}^{T} \frac{\partial L}{\partial z_t[i]} \frac{\partial z_t[i]}{\partial W_{xh}[i,j]} \quad \text{because } W_{xh} \text{ influences } L \text{ through each } z_t \\
&= \sum_{t=1}^{T} \frac{\partial L}{\partial z_t[i]} x_t[j] \\
&= \sum_{t=1}^{T} \delta_t[i] x_t[j]
\end{aligned}$$

$$\boxed{\frac{\partial L}{\partial W_{xh}} = \sum_{t=1}^{T} \delta_t x_t^T}$$

4.1.11 Calculation of $\frac{\partial L}{\partial W_{hh}}$

$$\begin{aligned}
\frac{\partial L}{\partial W_{hh}[i,j]} &= \sum_{t=1}^{T} \frac{\partial L}{\partial z_t[i]} \frac{\partial z_t[i]}{\partial W_{hh}[i,j]} \quad \text{because } W_{hh} \text{ influences } L \text{ through each } z_t \\
&= \sum_{t=1}^{T} \frac{\partial L}{\partial z_t[i]} h_{t-1}[j] \\
&= \sum_{t=1}^{T} \delta_t[i] h_{t-1}[j]
\end{aligned}$$

$$\boxed{\frac{\partial L}{\partial W_{hh}} = \sum_{t=1}^{T} \delta_t h_{t-1}^T}$$

4.1.12 Calculation of $\frac{\partial L}{\partial b_h}$

$$\begin{aligned}
\frac{\partial L}{\partial b_h[i]} &= \sum_{t=1}^{T} \frac{\partial L}{\partial z_t[i]} \frac{\partial z_t[i]}{\partial b_h[i]} && \text{because } b_h \text{ influences } L \text{ through each } z_t \\
&= \sum_{t=1}^{T} \frac{\partial L}{\partial z_t[i]} \\
&= \sum_{t=1}^{T} \delta_t[i]
\end{aligned}$$

$$\boxed{\frac{\partial L}{\partial b_h} = \sum_{t=1}^{T} \delta_t}$$

4.1.13 Calculation of $\frac{\partial L}{\partial b_y}$

$$\begin{aligned}
\frac{\partial L}{\partial b_y[i]} &= \sum_{t=1}^{T} \frac{\partial L_t}{\partial b_y[i]} && \text{because } b_y \text{ influences } L \text{ through each } L_t \\
&= \sum_{t=1}^{T} \frac{\partial L_t}{\partial y_t[i]} \frac{\partial y_t[i]}{\partial b_y[i]} \\
&= \sum_{t=1}^{T} \left(p_t[i] - y_t^{\text{true}}[i]\right)
\end{aligned}$$

$$\boxed{\frac{\partial L}{\partial b_y} = \sum_{t=1}^{T} \delta_t^y}$$

4.2 Backward Pass Algorithm (Backpropagation Through Time)

Initialize all parameter gradients to 0

For $t = T$:

$$
\begin{aligned}
\delta_T^y &= p_T - y_T^{\text{true}} \\
\frac{\partial L}{\partial W_{hy}} &+= \delta_T^y h_T^T \\
\frac{\partial L}{\partial b_y} &+= \delta_T^y \\
\frac{\partial L}{\partial h_T} &= W_{hy}^T \delta_T^y \\
\delta_T &= \frac{\partial L}{\partial h_T} \odot (1 - h_T \odot h_T) \\
\frac{\partial L}{\partial W_{xh}} &+= \delta_T x_T^T \\
\frac{\partial L}{\partial W_{hh}} &+= \delta_T h_{T-1}^T \\
\frac{\partial L}{\partial b_h} &+= \delta_T
\end{aligned}
$$

For $t = T-1, T-2, \ldots, 1$:

$$
\begin{aligned}
\delta_t^y &= p_t - y_t^{\text{true}} \\
\frac{\partial L}{\partial W_{hy}} &+= \delta_t^y h_t^T \\
\frac{\partial L}{\partial b_y} &+= \delta_t^y \\
\frac{\partial L}{\partial h_t} &= W_{hy}^T \delta_t^y + W_{hh}^T \delta_{t+1} \\
\delta_t &= \frac{\partial L}{\partial h_t} \odot (1 - h_t \odot h_t) \\
\frac{\partial L}{\partial W_{xh}} &+= \delta_t x_t^T \\
\frac{\partial L}{\partial W_{hh}} &+= \delta_t h_{t-1}^T \\
\frac{\partial L}{\partial b_h} &+= \delta_t
\end{aligned}
$$